

Aditya Singh

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Education

Arizona State University

Bachelor of Science, Computer Science

Aug 2022 - May 2026

Tempe, AZ

- **GPA:** 3.99/4.00
- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Organization & Assembly, Principles of Programming Languages, Foundations of Machine Learning, Intro. to A.I., Database Management, Distributed Software Development, Software Engineering, Information Assurance, Data Visualization, Exploring data in R and Python

Experience

Qualaces Inc.

Aug 2025 - May 2026

Software Engineering Intern, Core AI Team

Tempe, AZ

- Led development of a HIPAA-compliant medication management platform (LRWebApp) architected to scale for 500+ users, engineering an OCR-to-LLM extraction pipeline using PaddleOCR (replacing Tesseract.js, boosting accuracy from ~40% to 70%+) and a locally hosted SmolLM2 1.7B model via node-llama-cpp with hallucination filtering and JSON grammar constraints.
- Built the React/TypeScript frontend and Node.js/Express REST API, deployed cross-platform on Android, iOS, and web via Capacitor.js, with Supabase Auth, Row Level Security, and audit logging for HIPAA compliance on biweekly Agile sprints.

Attackfence Techlabs

Jun 2025 - Aug 2025

Software Engineering Intern

UP, India

- Engineered a full-stack Self-Evaluation Portal (DABProm) with Python, Streamlit, and MySQL, implementing secure bcrypt authentication with brute-force prevention and 4 role-based dashboards (Employee, Manager, HR, Admin).
- Reduced performance review completion time by 30% for managers and built a real-time notification system that increased user engagement by 25% across 20+ employees.

Aubot

Mar 2025 - May 2025

Software Engineering Intern

Remote

- Reviewed and enhanced 150+ Java and Python coding exercises, improving clarity and technical accuracy for over 1,000 students aged 5-17, contributing to a 15% improvement in comprehension rates.
- Collaborated with the content team to validate and refine instructional materials before Webflow integration, accelerating the development pipeline by 20%.

Projects

Lofty AI CRM (GlobeHack Winner, Lofty Track) (Next.js, TypeScript, Tailwind CSS, DeepSeek, Vapi, ElevenLabs, Google Maps)

2026

- Won the Lofty track at GlobeHack 2026 by building 5 AI-powered features on top of the Lofty real estate CRM: an outbound AI voice caller using Vapi and ElevenLabs that qualifies hot leads and books appointments to Google Calendar, a conversational AI control panel powered by DeepSeek, a Google Maps multi-stop route optimizer, automated pre-showing SMS alerts, and an AI-customizable dashboard.
- Engineered an SSE-streamed agentic onboarding flow with tool-calling that builds a 9-widget dashboard from freeform workflow descriptions in real time, plus an outreach pipeline that texts and emails high-intent leads and triggers outbound voice calls with structured-data extraction for automatic appointment booking.

MigrateMate (Python, FastAPI, Next.js, Docker, Neo4j, Redis, Qdrant)

2026

- Built an open-source tool automating Flask-to-FastAPI migration using an LLM and Python AST parsing, achieving more conversion accuracy across 20+ Flask patterns including Blueprint-to-APIRouter conversion, async route generation, and JWT auth translation.
- Architected a containerized backend via Docker Compose integrating 4 datastores (PostgreSQL, Redis, Qdrant, Neo4j) for context-aware retrieval-augmented generation (RAG), reducing migration review time against manual rewrites.

Library Management System (Node.js, Express, PostgreSQL, bcrypt)

2025

- Built a full-stack web application with full CRUD across 9 normalized tables, implementing relational schema design with foreign keys, multi-table joins, transactional borrowing logic, and bcrypt password hashing for secure authentication.
- Engineered pagination, real-time SQL ILIKE search across 3 entity types, and dynamic dropdowns with optimized queries that reduced average page render to under 200ms.

Data Analysis & Machine Learning Projects (Python, R, PyTorch, scikit-learn, TensorFlow)

2025

- Trained CNN image classifiers reaching higher accuracy, implemented a Deep Q-Learning agent in PyTorch, and built transformer attention from scratch; engineered a TF-IDF and cross-encoder NLP pipeline scoring <0.83 F1 on the Quora Question Pairs task.
- Performed logistic regression in R on a 400+ row health dataset and exploratory analysis in Python/pandas, producing 6+ visualizations and technical summaries for non-technical audiences.

Physical Therapy Landing Page (HTML, CSS, JavaScript)

2025

- Built an SEO-friendly landing page for a Luminous Rehab business, implementing sitemap.xml, robots.txt, structured metadata, and Open Graph/Twitter Card tags, achieving a Lighthouse SEO score of 100 and performance score of 95+.
- Deployed on Netlify with custom redirect rules and GitHub Pages fallback, featuring responsive design across 3 breakpoints and accessibility-focused markup scoring 100 on Lighthouse Accessibility.

Technical Skills

- **Languages:** Java, Python, C/C++, JavaScript, TypeScript, SQL, R, Go, Bash, HTML/CSS
- **Frontend & UI:** React.js, Next.js, Tailwind CSS, JavaFX, Streamlit, Capacitor.js, Radix UI, Bootstrap
- **Backend & Databases:** Node.js, Express.js, FastAPI, Spring Boot, PostgreSQL (Supabase), MySQL, MongoDB, Neo4j, Redis, Qdrant
- **AI & Data:** PyTorch, TensorFlow, scikit-learn, LLMs, LangGraph, OCR (PaddleOCR, Tesseract), RAG, CNNs, NLP, OpenCV, ETL Pipelines, Data Visualization
- **DevOps & Tools:** Git, Docker, Vercel, Render, CI/CD (GitHub Actions), Linux/Unix, Postman, Agile/Scrum